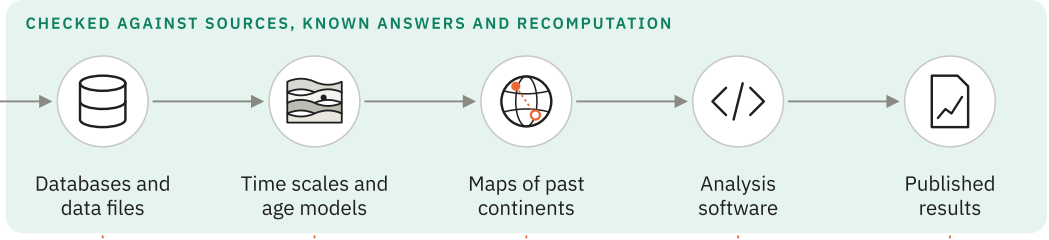
AI agents audit the shared foundations of fossil science

Twelve findings from one research program, each re-checked from fixed, checksummed inputs





two AI agents find, then re-check each from fixed inputs

1 in 5

mollusk records in one study's deposit absent from a dated public snapshot

460

interval ages that shifted between two database downloads

13.6°

median spread in ancient latitude of tested sites among three maps, 400 million years ago

3,972

of 8,282 layer positions given the wrong age in a test with known answers

3.4×10^{-16}

a printed probability; its own stated test gives 0.32

12 findings, each with its evidence and the command that reproduces it

across data, code, models and meaning

6 of the agents' own first accounts were corrected by the re-check

the auditors were audited too

WHAT RESEARCHERS CAN DO

- Record download dates and versions
- Compare more than one model
- Recompute a number before reuse